# EmailCheck — Email Verification Web Application

**Project Submission Document (v2.2)** | Genuine SMTP Probing, Render Port 25 Fallback & HTTPS Platform

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| **Submission Date:** October 26, 2026 | **Stack:** Python 3.11, Flask 3.1, dnspython, smtplib, HTTPS APIs, Vanilla JS, CSS3 |
| **Test Suite:** 65 / 65 Tests Passed (100% Pass Rate) | **Hosting Target:** Render Cloud Web Service (WSGI Gunicorn) |

## 1. Executive Project Overview

EmailCheck is a responsive web application designed for non-invasive email verification and deliverability diagnostics. Version 2.2 introduces an optional genuine SMTP recipient probing module (smtp_probe.py) that performs safe protocol handshakes (EHLO → MAIL FROM → RCPT TO → RSET → QUIT). To resolve cloud hosting restrictions where outbound TCP port 25 is restricted (such as on Render), the application integrates a safe circuit breaker that detects network blocks and seamlessly falls back to reputable verification APIs operating over secure HTTPS (port 443). Addresses are conservatively classified into **Valid**, **Invalid**, **Risky**, and **Unknown**.

## 2. Verified Live Application & Repository Links

| Item | Status | URL / Action Details |
|---|---|---|
| **GitHub Repository** | **Verified & Active** | Public GitHub Repository: https://github.com/Zenith150/emailcheck<br>Branch: main (Fully committed, verified, and synchronized). |
| **Live Cloud Deployment** | **Verified Live** | Production Deployed Application: https://emailcheck-7c9y.onrender.com/<br>Verified live on Render. Public endpoints: /, /health (200 OK, v2.2.0), and POST /api/verify. |
| **Production Health Status** | **200 OK (v2.2.0)** | Production endpoint: https://emailcheck-7c9y.onrender.com/health. Exposes version 2.2.0, active HTTPS provider status, and SMTP probing configuration. |

## 3. Transparent Classification Framework

| Classification | Evaluation Stage & Diagnostics | Meaning & Explanation |
|---|---|---|
| **Valid** | Valid RFC syntax; verified domain with active MX; not disposable; recipient inbox verified via SMTP 250 OK or HTTPS provider. | Syntax valid, active MX routing verified, and recipient inbox confirmed deliverable (High deliverability confidence). |
| **Invalid** | Malformed syntax, NXDOMAIN, RFC 7505 Null MX ('0 .'), missing mail routing, remote SMTP rejection (550), or undeliverable inbox. | Explicit diagnostic reason provided (e.g. malformed syntax, nonexistent domain, Null MX rejection, or undeliverable mailbox). |
| **Risky** | Known disposable domain, catch-all server (accepts all addresses), role account, or unverified inbox in Local Fallback mode. | Distinguishes domain routing from recipient inbox confirmation. Flags temporary inboxes and unverified mailboxes. |
| **Unknown** | DNS query timeout, temporary SMTP failure (450 Greylisting), provider API rate limit (429), or host port 25 block. | Transparent fallback: Never assumes deliverability when diagnostics or network connections are inconclusive. |

## 4. Genuine SMTP Probing & Solution to Render Port 25 Restriction

**Render Network Restriction:** Cloud platforms including Render block outbound TCP traffic on port 25 to prevent spam abuse. Port redirecting to 587/465 is impossible for recipient verification because submission ports require authentication credentials.
**Genuine SMTP Probe Module:** smtp_probe.py implements a protocol-compliant recipient handshake (EHLO → MAIL FROM → RCPT TO → RSET → QUIT). It strictly adheres to anti-spam ethics: **NEVER issues DATA**, **NEVER sends email messages**, and rejects RFC 7505 Null MX before connecting.
**Intelligent Circuit Breaker & Fallback:** In restricted environments (Render), the first port 25 timeout immediately trips an in-memory circuit breaker, cleanly marking remaining addresses as unavailable and falling back to HTTPS verification providers (port 443). This completely eliminates hanging requests and guarantees fast, reliable batch processing.

## 5. Application Architecture & Implemented Components

| Component | Implementation Details |
|---|---|
| **7-Stage Pipeline** | verifier.py provides syntax parsing, normalization, disposable domain filtering (200+ domains), DNS resolution, RFC 7505 Null MX detection, HTTPS provider verification, and classification. |
| **SMTP Probe Module** | smtp_probe.py provides safe, genuine SMTP probing, Null MX rejection, catch-all detection, response sanitization, and circuit breaker fallback for restricted hosts like Render. |

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| **Modular Providers** | providers.py implements BaseVerificationProvider with AbstractAPIProvider, HunterProvider, ZeroBounceProvider, and LocalFallbackProvider. |
| **User Interface** | Responsive dashboard, real-time address counter, 5 summary metric cards, status filters, instant search, pagination, collapsible diagnostics, Domain Diagnostics tool, Verification History, and RFC 4180 CSV export. |
| **Security & Privacy** | Server-side API keys, zero email transmission (no DATA command), zero database persistence, anonymized logging, sliding-window IP rate limiter, and XSS sanitization. |

## 6. Automated Test Suite Summary

The automated test suite across tests/test_verifier.py, tests/test_smtp_probe.py, and tests/test_api.py was executed and passed completely with **65 passed tests in 0.35 seconds**:

• Authoritative 7-Stage Pipeline definition & regression verification against LaTeX leaks: **PASSED**
• RFC 5322 Syntax validation & Local-part preserving Normalization: **PASSED**
• Disposable domain detection (Mailinator, Sharklasers, GuerrillaMail, etc.): **PASSED**
• DNS MX resolution, NXDOMAIN detection, RFC 7505 Null MX ('0 .'), and RFC 5321 implicit MX: **PASSED**
• Genuine SMTP probing protocol exchanges (250 OK, 550 Rejection, 450 Greylisting, Catch-all): **PASSED**
• SMTP Port 25 circuit breaker tripping, Null MX pre-connection abort, and DATA prevention: **PASSED**
• Opt-in checkbox request extraction (check_smtp & smtp_check) and server security controls: **PASSED**
• Server-disabled reporting vs. host port 25 block reporting and UI rendering regression: **PASSED**
• HTTPS Provider integrations (Abstract, Hunter, ZeroBounce, Local fallback): **PASSED**
• Classification logic & clearly explaining recipient verification availability: **PASSED**
• Batch memoization, 100-email size boundaries, rate limiting, and API error handlers: **PASSED**

**Privacy & Deliverability Guarantee:** EmailCheck guarantees that no emails are sent at any time. Verification operates purely over non-invasive structural diagnostics, safe protocol handshakes (without DATA), and secure HTTPS APIs (port 443). User lists are processed in-memory and are never permanently stored.